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Leaky Autofill: An Empirical Study on the Privacy Threat of Password Managers' Autofill Functionality

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Passwords suffer from usability-security dilemma

Increasing account numbers

80-112 accounts[1-2] before COVID-19

↓

168 accounts[3] after COVID-19

Password guidelines

Creating **random & unique** passwords for each account

↓

Enhancing password security for web users

Heavy management burden!

Using **popular or similar** passwords across multiple accounts

↓

Passwords are reused and guessable

[1] Leveraging semantic transformation to investigate password habits and their causes. In Proc. CHI 2018, pp. 1-12.
[2] Let's Go in for a Closer Look: Observing Passwords in Their Natural Habitat. In Proc. ACM CCS 2017, pp. 295-310.
[3] Juggling security: How many passwords does the average person have in 2024? https://nordpass.com/blog/how-many-passwords-does-average-person-have/

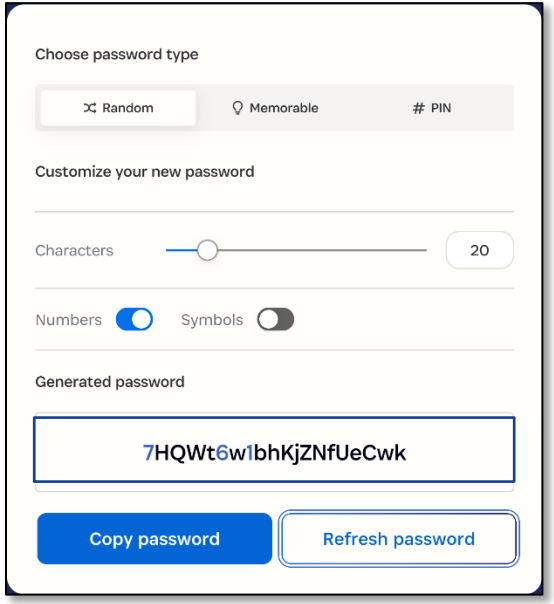
Password managers provide a technical solution

Built-in-browser PMs

LastPass
bitwarden
RoboForm
Sticky Password
securing your personal data
1Password
KEEPER
kaspersky
AVIRA
True Key
Proton Pass
DASHLANE
PASSWORD BOSS
norton

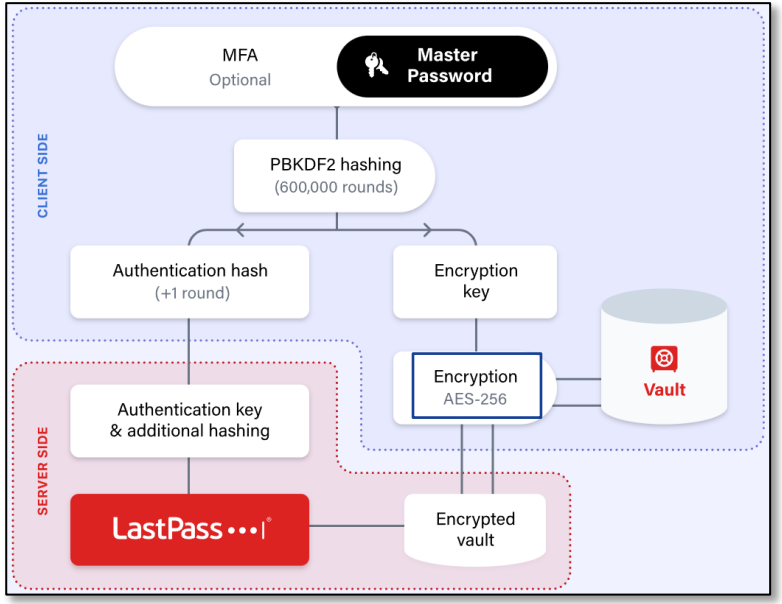
Separately-installed PMs

Password managers provide a technical solution



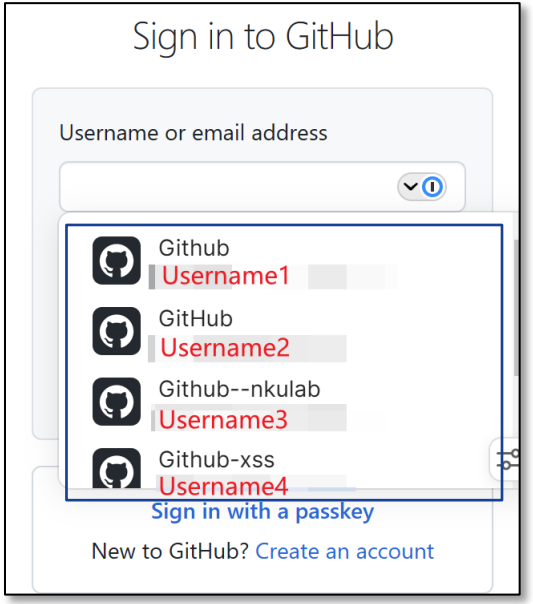
[1Password]

Generate strong passwords



[LastPass]

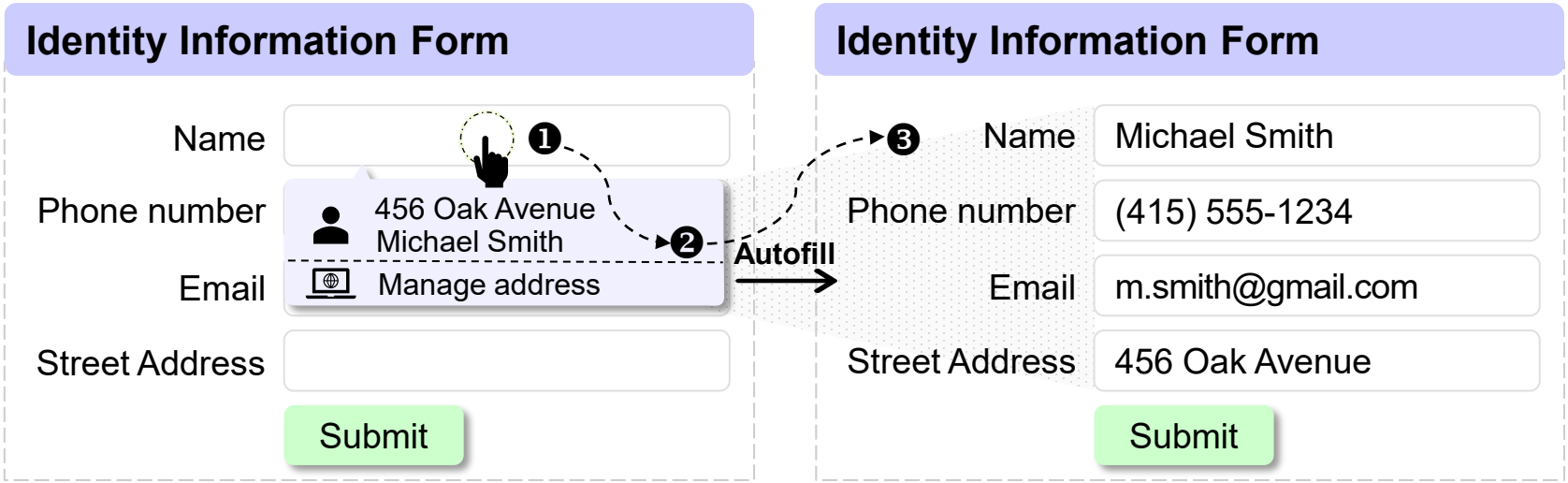
Encrypted storage



[1Password and GitHub]

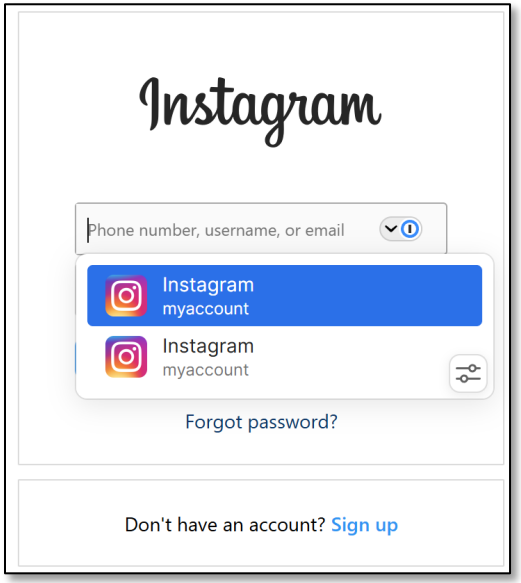
Convenient autofill

Introduction to the autofill functionality

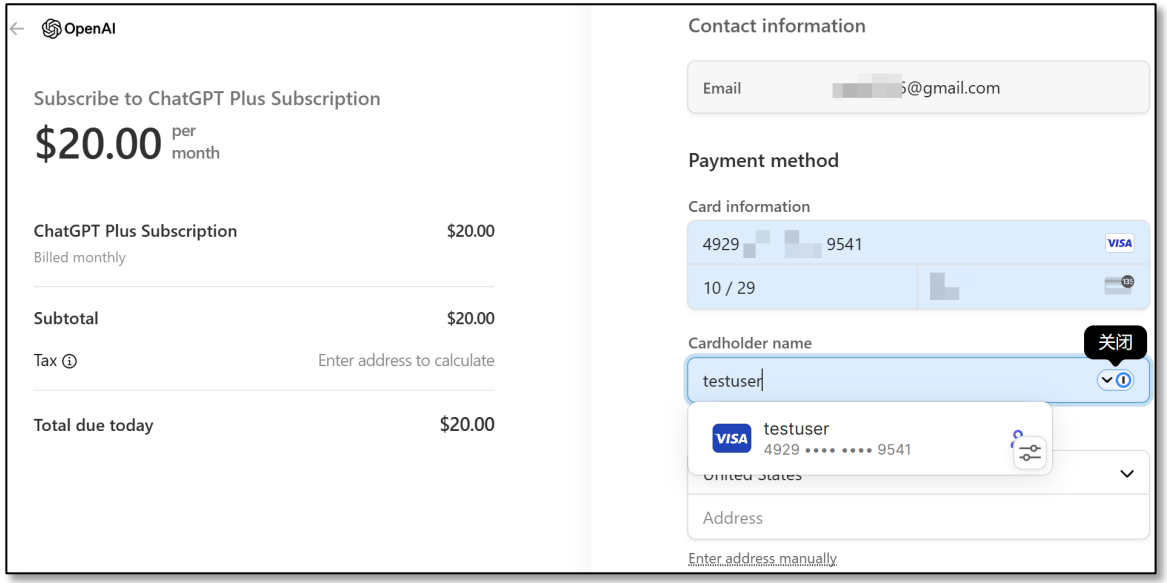


Users only need **one or a few** click(s) to fill in the web form with the autofill functionality

Introduction to the autofill functionality



[1Password and Instagram]

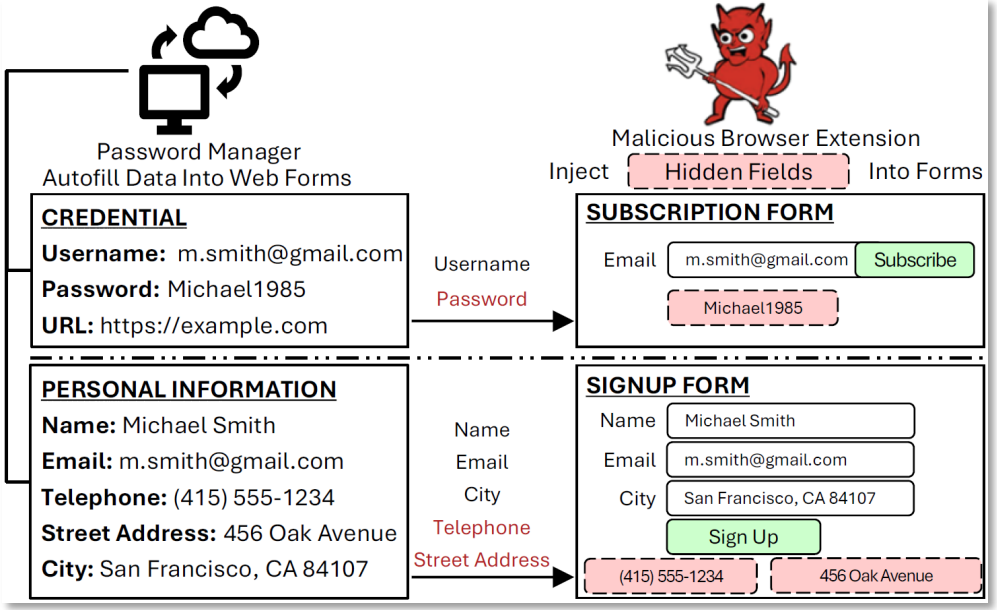


[1Password and Open AI]

Users only need **one or a few** click(s) to fill in the web form with the autofill functionality

Problem statement

An attack example



Threat model

The attacker could:

(1) … inject **invisible** <input> elements into web forms, e.g., password, address

(2) … wait for the user triggering the autofill functionality to fill in stored data

(3) … retrieve the filled **sensitive** data without users' **knowledge** or **consent**

Type of attackers:

(1) Malicious browser extensions

(2) Curiosity-driven websites

Our work

Previous research

Source	Target Object	Concealment Techniques
ACM CCS'20[4]	Six built-in-browser PMs	Eight concealment techniques
PETS'20[5]	Only Firefox browser	Not specified
ASIACCS'14[6]	Six built-in-browser PMs	Only `hidden`

- Only built-in-browser password managers
- Limited concealment techniques
- **A question arises:**

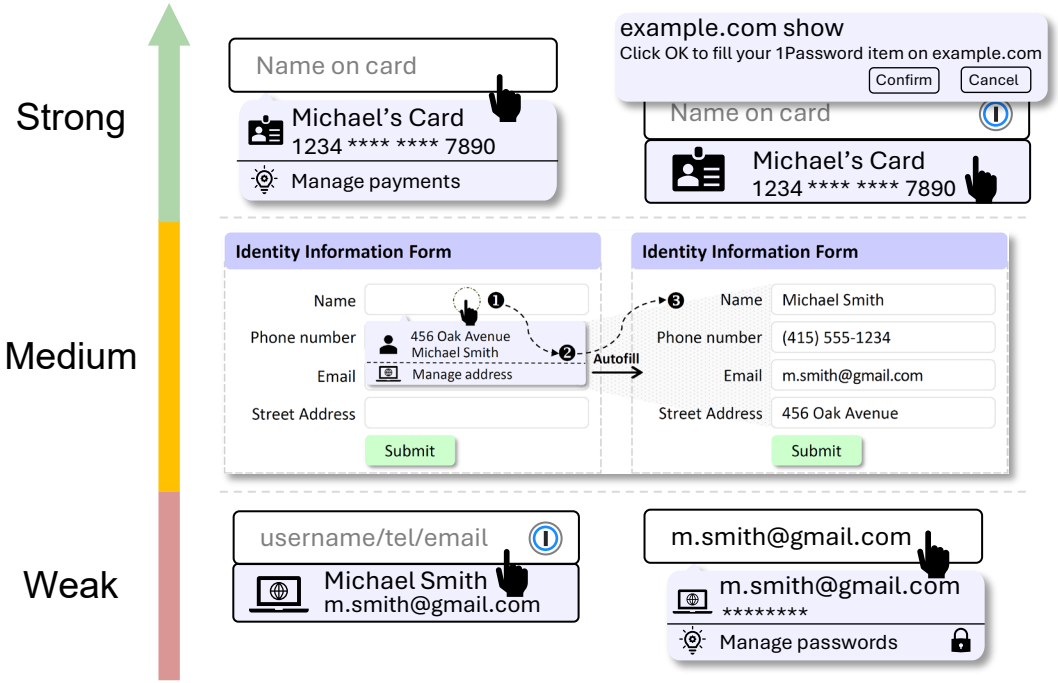
How do perceived more secure **separately-installed password managers** perform on identifying invisible fields on web forms?

Overview of our work

- **30** password managers, including **24** separately-installed and **six** built-in-browser password managers
- **15** concealment techniques for web form fields, with **seven** newly considered techniques, such as `clip`, `clip-path`, `transform`, `content-visibility`, …
- **Three** kinds of web forms:
  - Identity information form
  - Credit card form
  - Login form

[4] Fill in the Blanks: Empirical Analysis of the Privacy Threats of Browser Form Autofill. In Proc. ACM CCS 2018, pp. 507-519.
[5] No boundaries: data exfiltration by third parties embedded on web pages. In Proc. PETS, pp. 295-310.
[6] Protecting Users Against XSS-based Password Manager Abuse. In Proc. AsiaCCS, pp. 183-194.

RQ1: User interaction strength of the autofill functionality - Intro



Strong user interaction strength

- Detailed information of filled data and form
- Warnings or re-authentication for users

Medium user interaction strength

- Require **user interaction** to trigger the autofill functionality, yet provide **unclear** information about the filled data type

Weak user interaction strength

- Provide nothing about form and data type
- Autofill on page loading

RQ1: User interaction strength of the autofill functionality - Result

TABLE 1: Selected password managers, default autofill behaviors, and the interaction strength*.

Name	Info†	Version	Personal Info				Credit Card				Login			
			Method‡	Detail	Prompt	Level♮	Method	Detail	Prompt	Level	Method	Detail	Prompt	Level
LastPass: Free Password Manager	10,000k	4.130.2.1	ClickIcon	◇	Warn	S	ClickIcon	✓	Warn	S	On load	✓	⊘	W
Avira Password Manager	6,162k	2.20.0.4570	-	-	-	-	-	-	-	-	On load	✓	⊘	W
Norton Password Manager	5,194k	8.2.0.161	-	-	-	-	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S
1Password – Password Manager	4,443k	2.23.3	ClickIcon	×	⊘	W	ClickIcon	✓	Warn	S	ClickIcon	×	⊘	W
Bitwarden - Free Password Manager	3,903k	2024.4.1	RightClick	◇	⊘	M	RightClick	✓	⊘	S	ClickIcon	×	⊘	W
Kaspersky Password Manager	2,385k	24.0.128.1	ClickIcon[1]	✓	⊘	S	ClickIcon[1]	✓	Warn	S	On load	×	⊘	W
Dashlane — Password Manager	2,194k	6.2418.0	ClickIcon	◇	⊘	M	ClickIcon	✓	Mpw	S	On load	✓	⊘	W
iCloud Passwords	2,035k	2.2.9	-	-	-	-	-	-	-	-	ClickIcon	✓	⊘	S
Keeper Password Manager & Digital Vault	1,343k	16.8.3	RightClick	◇	Warn	S	RightClick	✓	Warn	S	ClickIcon	✓	⊘	S
MultiPassword — Password manager	1,288k	0.97.4	-	-	-	-	-	-	-	-	ClickIcon	×	⊘	W
True Key by McAfee	801k	4.3.1.9339	-	-	-	-	-	-	-	-	On load	×	⊘	W
RoboForm Password Manager	665k	9.5.9.2	ClickIcon	◇	⊘	M	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S
DualSafe Password Manager & Digital Vault	494k	1.4.28	-	-	-	-	-	-	-	-	On load	×	⊘	W
NordPass (desktop app version)	460k	5.15.28	ClickIcon	◇	⊘	M	ClickIcon[1]	✓	⊘	S	ClickIcon	✓	⊘	S
ExpressVPN Keys: Password Manager	391k	2.0.12.715	-	-	-	-	ClickIcon	✓	⊘	S	ClickIcon	×	⊘	W
Dropbox Passwords	374k	3.26.0	-	-	-	-	ClickIcon	✓	⊘	S	On load	✓	⊘	W
KeePassXC-Browser	369k	1.9.0.4	-	-	-	-	-	-	-	-	ClickIcon	×	Warn	S
NordPass Password Manager & Digital Vault	239k	5.15.29	ClickIcon	◇	⊘	M	ClickIcon[1]	✓	⊘	S	ClickIcon	✓	⊘	S
Passbolt - Open source password manager	233k	4.7.7	-	-	-	-	-	-	-	-	ClickIcon	✓	Mpw	S
Proton Pass: Free Password Manager	210k	1.14.1	-	-	-	-	-	-	-	-	ClickIcon	✓	⊘	S
Microsoft Autofill	140k	2.0.5	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S
Zoho Vault	134k	4.0	-	-	-	-	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S
Enpass Password Manager	124k	6.9.3	RightClick	◇	⊘	M	ClickIcon	✓	⊘	S	ClickIcon	×	⊘	W
Password Manager SafeInCloud	107k	24.1.0	-	-	-	-	Extension	✓	⊘	S	Extension	×	⊘	W
Google Chrome	65.38%	124.0.6367.119	ClickIcon	◇	⊘	M	ClickIcon	✓	⊘	S	On load	✓	⊘	W
Microsoft Edge	12.75%	123.0.2420.81	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S	On load	✓	⊘	W
Safari	8.72%	17.3.1	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S
Mozilla Firefox	7.26%	125.0.3	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S
Opera	3.05%	109.0.5097.80	ClickIcon	◇	⊘	M	ClickIcon	✓	⊘	S	On load	✓	⊘	W
Brave	-	1.65.130	ClickIcon	◇	⊘	M	ClickIcon	✓	⊘	S	ClickIcon	✓	⊘	S

* '-': Not applicable due to not being autofillable; ✓: Clear indication of filled form and data type; ◇: Only indication of filled form type; ×: No indication of what form and data will be filled; 'Warn': Warning dialog pops up before filling the form; 'Mpw':Master password required before filling the form; ⊘: No warnings shown, or permission and re-authentication required.
† Active users of Chrome browser extensions for 24 separately-installed PMs from ChromeStats [11] and market share of six browsers with built-in PMs sourced from StatCounter [56].
‡ Autofill triggering method. 'On load' means that the information is filled into the fields when the web page loads; 'ClickIcon' means users need to click the PM icon in the web form field or click the field to trigger the autofill functionality; 'ClickIcon[1]' means that the PM icon only appears in the *targeted sensitive* field; 'RightClick' means that users need to right-click the web page to select the PM menu to trigger the autofill functionality; 'Extension' means that users need to click the extension in the browser menu bar to trigger the autofill functionality.
♮ **W** means the PM has **W**eak user interaction strength for autofill functionality in this form, **M** for **M**edium interaction strength, and **S** for **S**trong interaction strength.

30 PMs autofill **69** scenarios

17 for personal information forms

- **10** PMs provide little information about filled data and form type

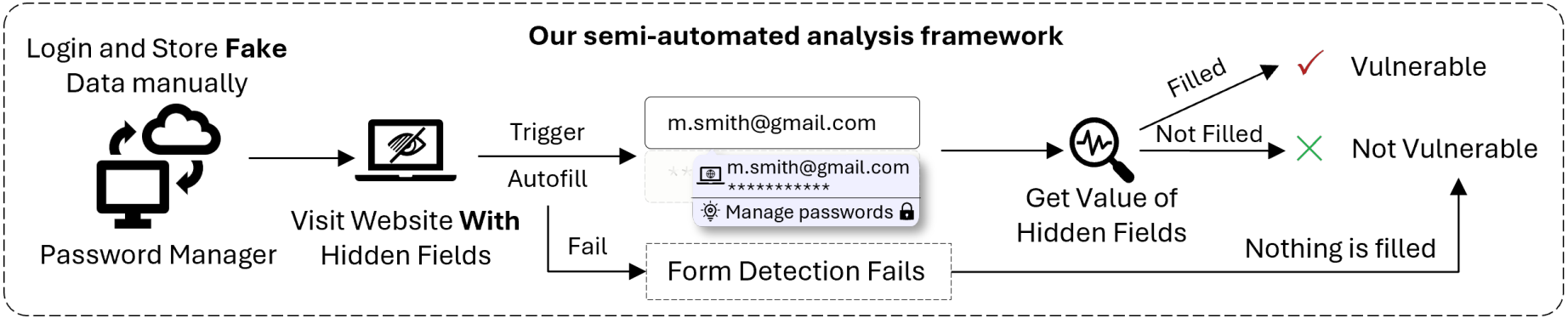
22 for credit card forms

- All PMs provide **strong** interaction

30 for login forms

- **16** PMs provide weak interaction
- **10** PMs autofill on page loading

RQ2: The ability in detecting hidden form fields - Method



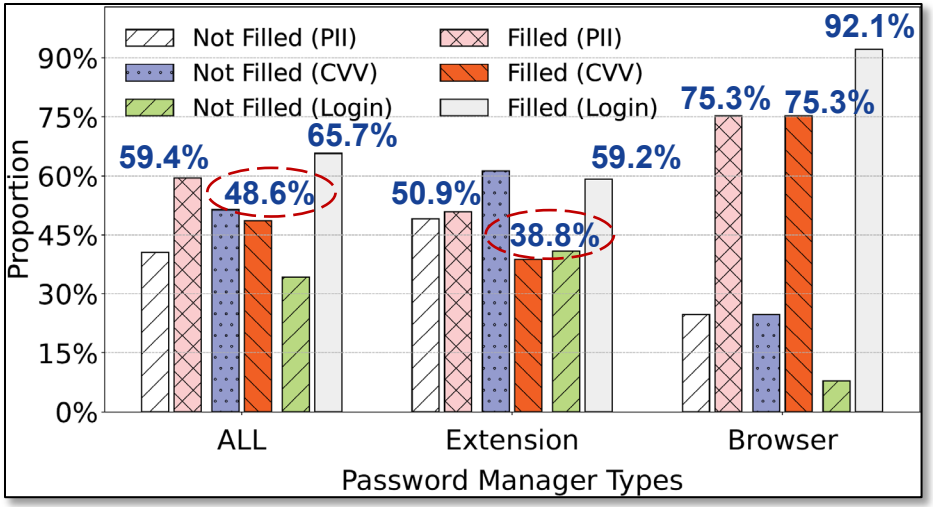
1. Start a browser instance and log into the password manager account
2. Import/Add test data into password manager for each web form

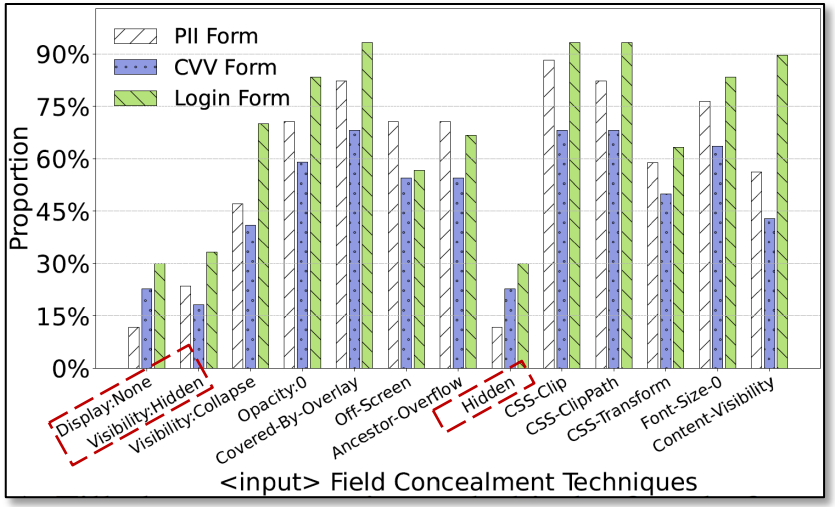
Require **one-time** human operation

3. Access the test website and trigger the autofill functionality using Selenium
4. Using Selenium to log which data gets auto-filled

Automated testing and logging using Selenium

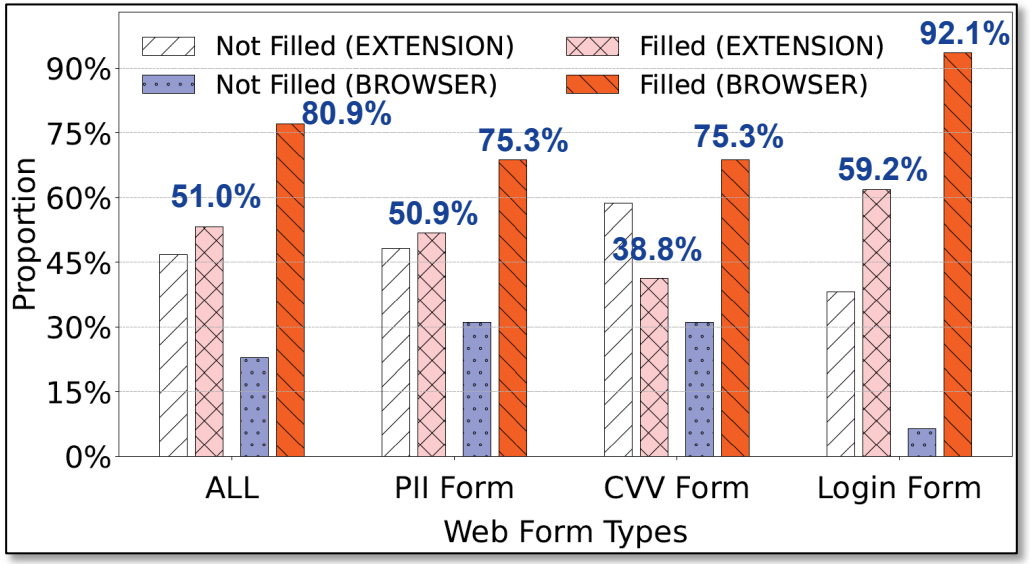
RQ2: The ability in detecting hidden form fields - Result





- Filled probability of **credit card forms** is **48.6%**, where **38.8%** for separately-installed PMs
- Hidden fields in **credit card forms** are significantly less likely (**0.494 times**) to be filled than **login forms**
- **83.3% (25/30)** PMs successfully detect **three techniques** in at least one web form

RQ3: Comparison between two kinds of password managers



- Built-in-browser PMs are more likely **(~4.07 times)** to fill data into hidden fields than separately-installed PMs
- No obvious improvements for built-in-browser PMs

Concealment Tech.†	Display: None	Visibility: Hidden	Visibility: Collapse	Opacity: 0	Covered by Overlay	Non-Effective-Size	Off-Screen	Ancestor-Overflow	Hidden	CSS Clip	CSS Clip-Path	CSS Transform	Font Size: 0	Content-Visibility	Tiny-Size
Chrome	×	×	×	✓	✓	✓	✓	✓	×	✓	✓	✓	✓	✓	✓
Edge	×	×	×	✓	✓	✓	✓	✓	×	✓	✓	✓	✓	✓	✓
Safari	×	×	×	✓	✓	×	✓	✓	×	✓	✓	✓	✓	-[1]	×
Firefox	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Opera	×	×	×	✓	✓	✓	✓	✓	×	✓	✓	✓	✓	✓	✓
Brave	×	×	×	✓	✓	✓	✓	✓	×	✓	✓	✓	✓	✓	✓

Table 1: Browsers that autofill form fields that are hidden from the user, based on various concealment techniques.

Techniques	Firefox	Chrome	Brave	Edge	Safari	Opera
CSS Display	✓	✗	✗	✗	✗	✗
CSS Visibility	✓	✗	✗	✗	✗	✗
CSS Opacity	✓	✓	✓	✓	✓	✓
Covered by overlay	✓	✓	✓	✓	✓	✓
Non-effective size	✓	✓	✓	✓	✓	✓
Off-screen placement	✓	✓	✓	✓	✓	✓
Ancestor's overflow	✓	✓	✓	✓	✓	✓

[From ACM CCS'20[4]]

[4] Fill in the Blanks: Empirical Analysis of the Privacy Threats of Browser Form Autofill. In Proc. ACM CCS 2018, pp. 507-519.

Issues report and countermeasures

Reporting privacy threats to password vendors

LastPass → **Confirmed** but not solved

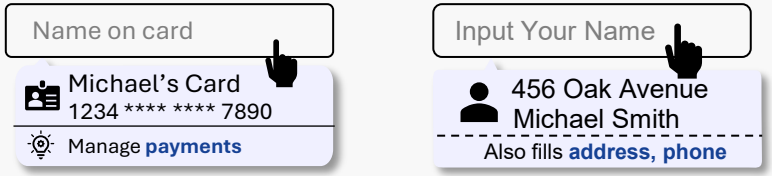
1Password → A compromised client-side is **not** considered

bitwarden, KeePassXC → Consider the **balance** between user experience, performance, and security

Two suggestions to reduce privacy threats

1. Increasing user interaction strength

- Prevent autofill on page loading
- Specifying the type of filled data and type



2. Using visual language model

- Counting and filling visible form fields

Input: *How many visible fields in this web form?*

Web form screenshots

→ VLM → **Output:** *Three visible input fields and one button*

Thank you!

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https://zenodo.org/records/13380735

https://github.com/Leaky-Autofill

https://leakyautofill.github.io/

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- An empirical study on the privacy threat of the **autofill functionality** of 30 password managers
- A **semi-automated** password manager autofill functionality end-to-end testing tool

More research about identity authentication security, see https://wangdingg.weebly.com/publications.html